AI CERTs™

AI+ Foundation™

Certification



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Module 2

Prompt Engineering - Interacting with AI

Overview

Prompt Engineering is a crucial skill in the effective interaction with AI models, enabling users to design inputs that generate the most accurate and relevant outputs. This course offers a comprehensive introduction to the principles and techniques of creating well-structured prompts for AI models. Using tools like ChatGPT, DALL-E, GPT, and other generative AI models, participants will explore methods to refine AI-generated responses, minimize biases, and maximize AI performance. The course provides hands-on experience through exercises and case studies, enabling learners to apply prompt engineering techniques in real-world scenarios across business, education, and creative industries. By the end of this course, participants will be equipped to structure prompts effectively to enhance AI's capabilities and drive impactful results.

This module covers:

- **Defining Prompt Engineering and Its Importance**: Understanding the fundamental concept of prompt engineering, its role in refining AI responses, and its significance in optimizing AI interactions.
- **Enhancement Through Well-Crafted Prompts**: Exploring how well-structured prompts can improve the quality of AI-generated responses, minimizing biases and increasing the relevance and accuracy of the outputs.

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- **Exploring GPT**: Advancements in GPT, including its improved context understanding, nuanced response generation, expanded knowledge base, and enhanced language coherence for diverse applications.
- **Use-Case Exploration**: Examining the application of various prompting techniques across different sectors, highlighting how prompt engineering is used in real-world scenarios to solve industry-specific challenges.
- **Activity: Hands-On - Writing Prompts**: Engaging in practical exercises on writing effective prompts, giving direction, formatting responses, providing examples, and evaluating AI-generated quality. Additionally, dividing labor for different AI tasks.
- **Activity: Hands-On - Discover How GPT Transforms Industries**: Practical demonstrations illustrating how GPT revolutionizes industries like business and healthcare through prompt engineering and AI-driven solutions.

2.1 Defining Prompt Engineering and its Importance

2.1.1 Definition

Prompt Engineering is designing and refining inputs (prompts) to guide AI models, such as GPT, in generating desired outputs. It involves structuring questions, commands, or requests to enhance the AI's understanding and performance, ensuring that the generated responses are accurate, relevant, and aligned with specific goals. Practical, prompt engineering helps optimize AI interactions, reduce biases, and improve the overall quality of AI-generated content. It is a crucial skill for utilizing generative AI models in various applications.

2.1.2 Importance of Prompt Engineering:

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- **Improves AI Output Quality**: Well-crafted prompts help generate more accurate, relevant, and coherent responses from AI models, enhancing the quality of the output.
- **Reduces Bias and Errors**: Effective, prompt engineering minimizes the risk of biased or incorrect responses by guiding the AI in a specific direction and setting clear expectations.
- **Optimizes AI Efficiency**: Properly structured prompts can make AI models more efficient, saving time by reducing the need for multiple iterations and refining responses.
- **Enhances Customization**: Prompt engineering allows users to tailor AI responses to meet specific needs across domains such as business, healthcare, or education, ensuring that the output aligns with the context.
- **Enables Complex Tasks**: By strategically crafting prompts, complex tasks can be broken down into manageable steps, allowing AI to tackle multifaceted problems more effectively.
- **Improves User-AI Interaction**: Good prompt engineering ensures smoother, more intuitive interactions between users and AI, enhancing the overall user experience.
- **Boosts AI Adoption Across Industries**: As industries adopt AI tools, prompt engineering plays a key role in ensuring AI is used effectively to address real-world challenges and provide valuable insights.
- **Facilitates Learning and Adaptation**: Effective prompts help AI models learn more effectively from data, improving their ability to adapt to new situations and generate innovative solutions.

2.2 Enhancement through Well-Crafted Prompts

Well-crafted prompts significantly enhance AI-driven statistical analysis by ensuring precise, relevant, and context-aware outputs. Thoughtfully structured prompts help extract deeper

insights, refine predictive models, and automate complex BI tasks efficiently. By leveraging clear, targeted queries, organizations can maximize AI's potential, improving decision-making, forecasting accuracy, and data interpretation in a scalable and intelligent manner.

Below are important aspects for enhancing AI-driven statistical analysis:

Aspect	Description	Importance	Example
Clear Objective Setting	Starting with a clear goal for the prompt helps in formulating a targeted question or request, ensuring AI generates responses aligned with your desired outcome.	Ensures focused and relevant responses, making the AI's output more aligned with the user's needs.	**Task**: Generate a list of benefits of AI in healthcare. **Prompt**: "Provide a detailed list of benefits of AI in healthcare, focusing on diagnostics, patient care, and cost reduction."
Specificity in Prompts	Being specific in the prompt prevents AI from	Specific prompts lead to more relevant, accurate,	**Task**: Generate a summary of a financial report.

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	generating broad or irrelevant responses, guiding it to provide more accurate, actionable answers.	and actionable results, saving time and effort.	**Prompt**: "Summarize the key financial highlights from Q1 2023, including revenue growth, expense management, and profit margins."
Contextualizing Prompts	Providing context helps AI better understand the prompt, improving its accuracy. However, balancing the amount of context is important to avoid overwhelming the system.	Context aids in delivering more accurate and relevant responses, while avoiding confusion caused by excessive context.	**Task**: Explain the concept of blockchain. **Prompt**: "Explain blockchain technology and its key components, focusing on decentralized networks and

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			their application in cryptocurrency.”
Balancing Specificity and Open-Endedness	Striking the right balance between specific and open-ended prompts allows for precise answers while also encouraging exploration and creativity.	Enables both targeted responses and creative exploration, expanding the range of possible outcomes.	**Task**: Generate creative ideas for a marketing campaign. **Prompt**: “Provide 3 creative campaign ideas for launching an eco-friendly product in the fashion industry. Include themes and possible slogans.”
Iterative Refinement of Prompts	Crafting effective prompts is an iterative process. Adjusting prompts based on AI feedback allows	Continuous refinement improves response quality, ensuring outputs are more aligned	**Task**: Generate a report on AI applications in education.

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	for fine-tuning responses to better meet objectives.	with the desired outcome.	**Prompt**: "Write a report about AI in education, focusing on personalized learning. Revise it to add real-world applications."
Analyzing AI Responses	Evaluate if the AI's output meets your expectations or if there are areas of ambiguity or misinterpretation.	Helps assess the relevance, clarity, and accuracy of AI's response, providing insights for further refinement.	**Task**: Explain climate change impacts. **Prompt**: "Describe the impact of climate change on agriculture and assess whether the AI's response included both environmental and economic impacts."

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Adjusting the Prompt	Modify the prompt based on feedback to make it more specific or provide additional context, improving the quality of AI-generated responses.	Fine-tuning the prompt improves the relevance and quality of AI responses, ensuring better alignment with the objective.	**Task**: Generate a product description. **Prompt**: "Write a product description for a new eco-friendly cleaning product. Adjust it to focus more on its environmental benefits after reviewing the first draft."
Re-evaluating Outputs	After adjusting the prompt, assess the new responses and repeat the cycle for continuous improvement.	Ensures that the final output is refined and aligns with the goal, allowing for continuous	**Task**: Create a detailed marketing plan. **Prompt**: "Create a marketing plan for an AI product.

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		optimization of AI's performance.	Reevaluate the strategy to focus more on digital channels after the first review."
Precision and Creativity	Effective prompt crafting involves both precision for exact responses and creativity to allow AI to generate diverse ideas or solutions based on open-ended queries.	Balances the need for targeted, specific answers with the freedom for creative exploration, resulting in innovative solutions.	**Task**: Generate a creative product idea. **Prompt**: "Generate a unique idea for a new wearable tech product for fitness enthusiasts. Focus on innovative features but keep it feasible for mass production."
Ongoing Improvement	The prompt engineering	Continuous adaptation and	**Task**: Analyze market trends.

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	process is dynamic, requiring continuous adaptation based on the quality of AI responses. Constant feedback and adjustment ensure optimal results.	feedback optimize AI responses over time, enhancing its performance and effectiveness.	**Prompt**: "Analyze the top five market trends for 2023 in the tech industry. Improve by adding trend forecasts for the next 5 years after assessing the first analysis."

2.3 Exploring GPT

GPT is a next-generation AI language model that significantly enhances natural language understanding, data analysis, and decision-making across diverse domains, providing more context-aware and nuanced responses compared to its predecessors. With advanced reasoning capabilities, it can process complex queries, generate insightful responses, and automate analytical tasks. In Business Intelligence and Statistical Analysis, GPT aids trend prediction, pattern recognition, and strategic forecasting, making data-driven insights more accessible and actionable. Its scalability and adaptability enable businesses to optimize workflows, improve efficiency, and make more informed decisions.

Below are the features of GPT, a powerful AI language model:

Feature	Description	Importance	Example
Enhanced Context Understanding	GPT has an enhanced ability to understand and retain context over longer conversations or larger inputs, making it more efficient in handling complex tasks.	This leads to more coherent and relevant responses, enhancing user experience in extended interactions.	In a long customer service chat, GPT can remember the user's previous issues and maintain a consistent conversation.
Nuanced Response Generation	GPT can generate more nuanced and sophisticated responses, noticing subtle details and nuances in language.	This allows it to provide more insightful and tailored answers compared to previous versions.	In a discussion about ethical dilemmas, GPT can provide deeper insights into moral reasoning, beyond simple answers.

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Expanded Knowledge Base	GPT has access to an even broader and up-to-date knowledge base, enabling it to provide more accurate, contextually relevant information across a wider range of topics.	This enhances its usefulness in diverse applications, making it more versatile in providing accurate information on a wide array of subjects.	GPT can provide more accurate medical information, referencing the latest research in fields like cancer treatment.
Improved Multimodal Capabilities	GPT produces more coherent and consistent responses, improving language flow and user engagement, even across long-form interactions	This ensures the flow of conversation remains logical, clear, and easy to follow, improving user engagement and experience.	In a complex academic explanation, GPT delivers clear and logically structured answers, ensuring readability.

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Diverse Applications	GPT's advancements in understanding context, generating nuanced responses, and improving coherence make it applicable across various domains like business, healthcare, education, and content creation.	These improvements expand their utility in solving real-world problems across multiple industries, enhancing productivity and decision-making.	GPT is used in business for generating marketing content, in healthcare for aiding diagnosis, and in education for tutoring.
Enhanced Handling of Complex Queries	GPT can manage complex or multifaceted queries more effectively, breaking them down into digestible components.	This ensures that GPT can provide detailed, comprehensive, and insightful answers, even for intricate or challenging queries.	When asked about a multi-step process like quantum computing, GPT provides an easy-to-understand breakdown of complex concepts.

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Better Multilingual Capabilities	GPT shows improved performance in multiple languages, making it more accessible for global applications and diverse user bases.	This increases its versatility and global accessibility, allowing it to serve a wider range of audiences in different languages.	GPT can translate and generate content fluently in languages like French, Spanish, and Chinese, offering accurate translations.
Reduced Biases and Errors	GPT works to minimize biases in its responses, improving fairness and accuracy in interactions.	This is essential for providing more objective and neutral answers, especially when dealing with sensitive or controversial topics.	When asked about sensitive topics, GPT generates fact-based responses, ensuring neutrality and minimizing bias in discussions.

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2.4 Use-case: Exploration of applying various prompting techniques across different sectors

2.4.1 Healthcare

- **Personalized Treatment Plans**: Using well-crafted prompts, AI can generate personalized, data-driven treatment recommendations based on real-time patient data, medical history, and predictive models.

  Significance: These tailored responses help create individualized treatment plans, leading to more effective care and better patient outcomes.
- **Medical Research**: Open-ended prompts to explore new drug formulations or potential treatment methods by leveraging vast medical databases and literature.

Significance: This helps accelerate research, discover innovative treatments, and provide valuable insights for advancing Healthcare.

2.4.2 Education

- **Adaptive Learning**: Specific prompts are important for tailoring educational content to the learner's needs.
- **Significance:** These prompts enhance engagement and improve learning outcomes by providing customized exercises and feedback based on learning styles.
- **Knowledge Assessment**: Structured prompts are essential for creating quizzes and assessments aligned with educational objectives.
- **Significance:** These prompts help ensure that evaluations are thorough and cover the necessary material, providing valuable insights into student performance.

2.4.3 Business and Marketing

- **Customer Insights**: Specific prompts for analyzing customer feedback, reviews, and behavior data to generate insights for improving products and services.

  Significance: These insights are critical for improving products and services, refining marketing strategies, and making data-driven decisions to meet customer needs better.
- **Content Creation**: Open-ended prompts enable the generation of creative marketing materials, blog posts, and social media content.

  Significance: These prompts allow businesses to create targeted content that resonates with specific audiences, enhancing brand engagement and marketing efforts.

2.4.4 Finance

- **Risk Analysis**: Structured prompts guide AI in extracting and analyzing financial data, helping identify potential risks, investment opportunities, and market trends.

  Significance: This is essential for informed decision-making and proactive risk management in the finance sector.
- **Fraud Detection**: Specific prompts enable AI to analyze transaction patterns and flag suspicious activities.

  Significance: This capability is crucial for detecting and preventing fraud in real-time, ensuring the security of financial institutions, and protecting their clients from potential losses.

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2.4.5 Creative Industries

- **Content Generation**: Open-ended prompts are key to unlocking creativity. They allow AI to generate artwork, designs, or stories based on user input.

  Significance: This enhances the creative ideation process and aids artists, writers, and designers explore innovative ideas and concepts.
- **Game Design**: Specific prompts generate elements of game design, such as character dialogue, plot ideas, or narrative arcs.

  Significance: This helps developers create engaging and immersive experiences, enhancing gameplay and storyline development.

2.4.6 Customer Service

- **Chatbots**: Well-structured prompts are vital for guiding AI-powered chatbots in delivering personalized and contextually accurate responses to customer inquiries.

  Significance: This improves customer satisfaction, speeds up response times, and enhances the user experience.
- **Automated Troubleshooting**: Specific prompts guide AI systems in diagnosing technical issues based on user input.

  Significance: This capability enables faster issue resolution, reduces the need for human intervention, and enhances the efficiency of customer support operations.

Activity 1

Writing prompts, Giving Direction, Formatting Responses, Providing Examples, Evaluating Quality, and Dividing Labor for different AI tasks.

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Activity	Description	Hands-on Example
Writing Prompts Giving Direction	Create clear and specific prompts that guide AI in producing desired outcomes for various tasks.	**Task**: Write a blog post about the benefits of AI in education. **Prompt**: "Write a 500-word blog post outlining the benefits of AI in education, focusing on personalized learning, adaptive learning platforms, and real-time feedback for students."
Formatting Responses	Design prompts that guide AI to format responses in a particular structure (e.g., bullet points, paragraphs).	**Task**: Summarize key benefits of using AI in customer service. **Prompt**: "Summarize the top five benefits of using AI in customer service. Please format your response in bullet points

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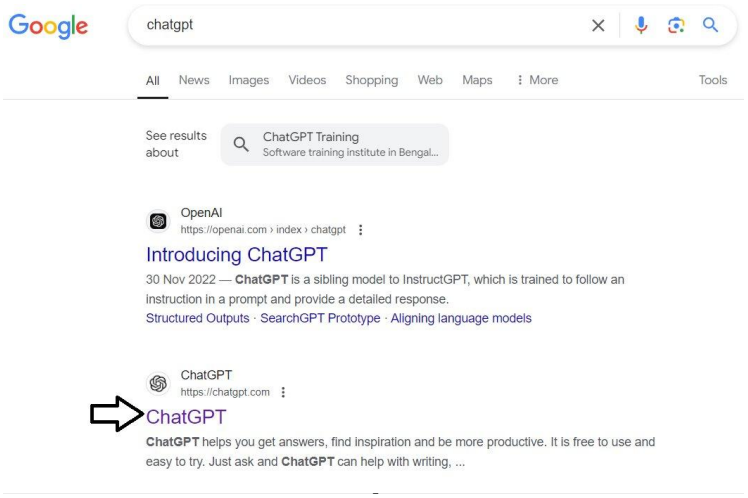
		and ensure each benefit is clearly explained.”
Providing Examples	Include examples in the prompts to help AI understand the context and generate more accurate responses.	**Task**: Provide a marketing strategy for a new AI product. **Prompt**: “Provide a marketing strategy for a new AI product aimed at the healthcare sector. For example, consider promoting its ability to improve patient care and streamline hospital operations.”
Evaluating Quality	Assess AI-generated responses based on clarity, relevance, accuracy, and coherence, and refine prompts accordingly.	**Task**: Generate a detailed description of AI in healthcare. **Prompt**: “Provide a detailed description of how AI is used in healthcare, with examples of its

		applications in diagnosis and treatment." **Evaluation**: After receiving the response, assess whether the information is clear, accurate, and relevant. If not, refine the prompt by adding more context or specifying areas that need to be covered.
Dividing Labor for Different AI Tasks	Develop prompts that assign specific tasks to AI, ensuring efficient division of work for diverse types of queries.	**Task**: Divide AI tasks between content creation and data analysis. **Prompt**: "First, summarize the main points of the article on AI in education. Then, analyze the data presented in the article to highlight trends and key insights. Please ensure to separate the summary

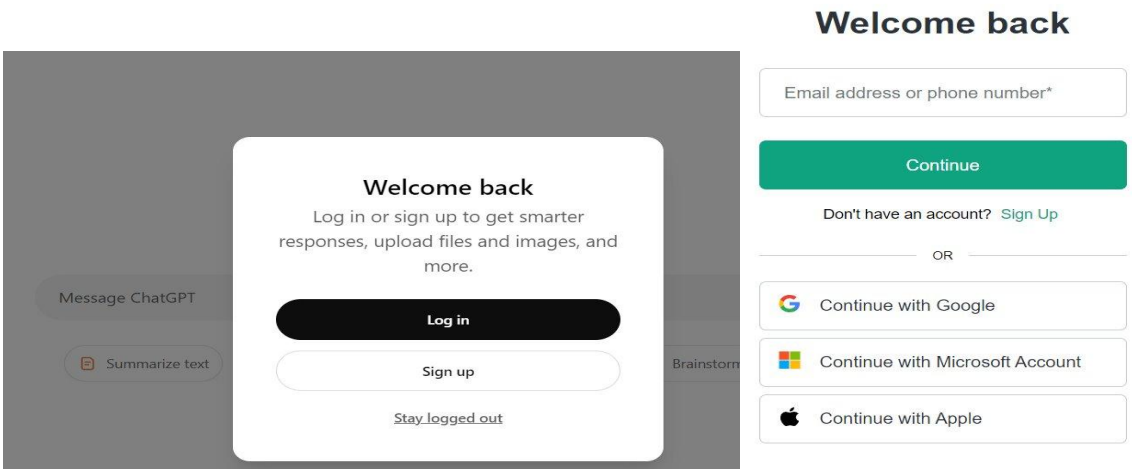
		and analysis into distinct sections.”

Applying the Five Principles

1. **Step-1:** Search ChatGPT

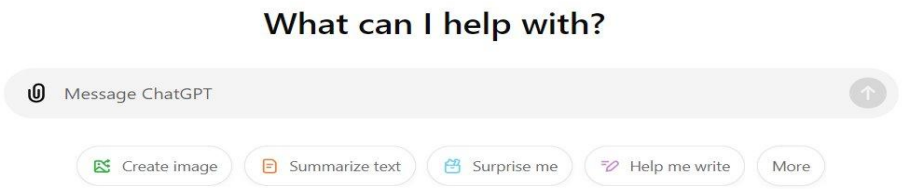


2. **Step-2:** Sign-up using Google id or Microsoft id or Apple id.



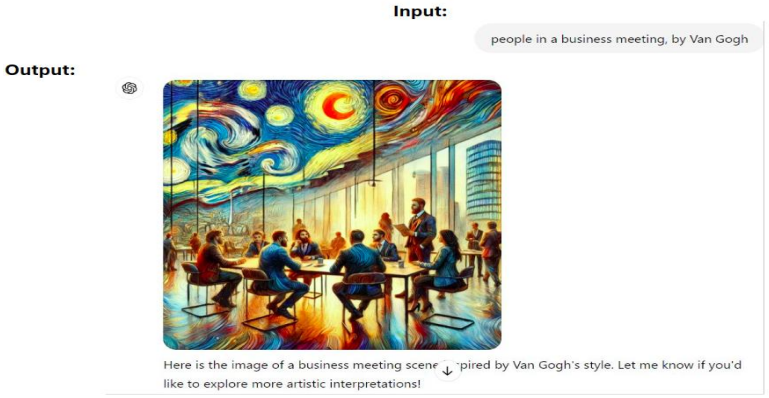
3. **Step-3:** Type your Prompt.

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Giving Direction

Example:



Try these prompts on Giving Direction

- Using this advice, can I have a list of product names for a pair of shoes that can fit any foot size?
- Rose flower, by any famous artists
- Give me exercises for leg pain for 50yrs people.

Formatting Responses

Example

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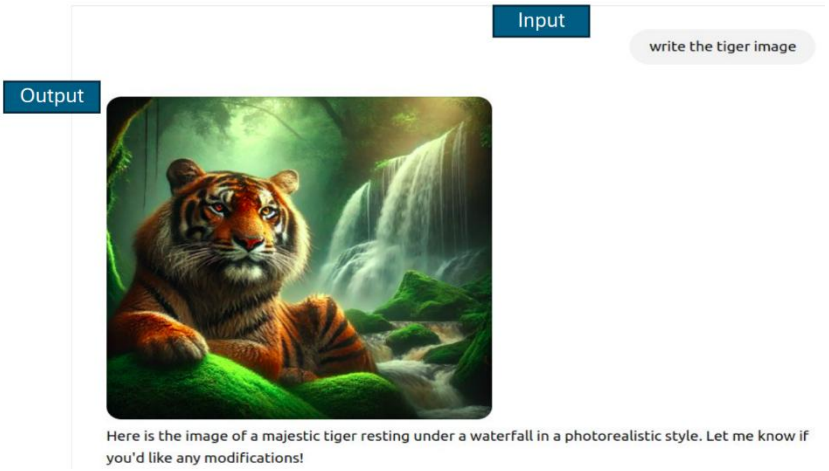


Try these prompts on Formatting Responses

Classroom study of four students watching MacBook on top of table, in London

Provide Examples

Examples



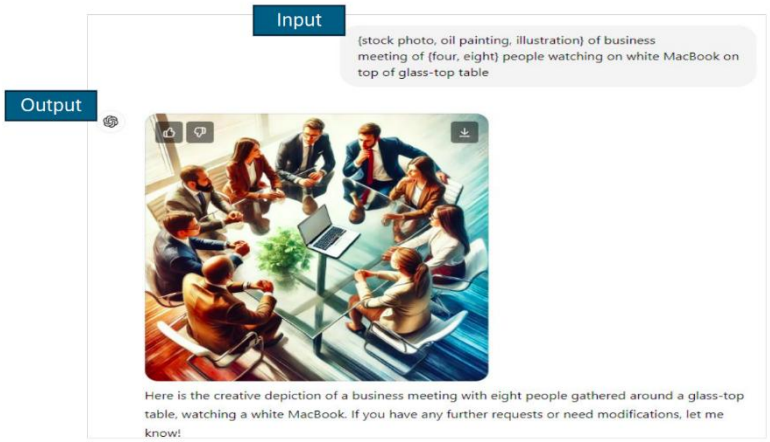
Try these prompts on Provide Examples

stock photo of business meeting of four people watching on white MacBook on top of glass-top table, Panasonic, DC-GH5

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Evaluating Quality

Example



Try these prompts on Evaluating Quality

{cricket bat photo, oil painting, illustration} of cricket ground of {two, four} people watching on white MacBook on top of glass-top table

Vs

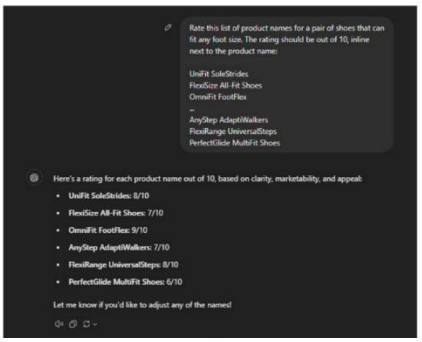
1. cricket bat photo of cricket ground of two people watching on white MacBook on top of glass-top table
2. cricket bat photo of cricket ground of eight people watching on white MacBook on top of glass-top table
3. cricket bat photo of cricket ground of four people watching on white MacBook on top of glass-top table
4. cricket bat photo of cricket ground of eight people watching on white MacBook on top of glass-top table
5. illustration of cricket ground of four people watching on white MacBook on top of glass-top table
6. illustration of cricket ground of eight people watching on white MacBook on top of glass-top table

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Dividing Labor

Example

Input:



Output:

Try these prompts on Dividing Labor

Let's think step by step. Rate this list of product names for a pair of shoes that can fit any foot size. The rating should be out of 10, inline next to the product name:

UniFit SoleStrides
FlexiSize All-Fit Shoes
OmniFit FootFlex
...
AnyStep AdaptiWalkers
FlexiRange UniversalSteps
PerfectGlide MultiFit Shoes

Activity 2

Discover how GPT transforms industries with practical demonstrations in business, healthcare, and various sectors.

Activity	Description	Hands-on Example

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Discover How GPT Transforms Business	Practical demonstration of how GPT aids data analysis, customer insights, and marketing strategies in business.	**Task**: Analyze customer reviews to identify key trends for improving a product. **Prompt**: "Summarize the key customer feedback trends from the reviews. Highlight common issues and feature requests to guide product improvement."
Discover How GPT Transforms Healthcare	Explore GPT's role in improving patient care through predictive analytics, medical research, and personalized treatment plans.	**Task**: Generate a personalized treatment plan based on patient data. **Prompt**: "Create a personalized treatment plan for a patient with diabetes, considering

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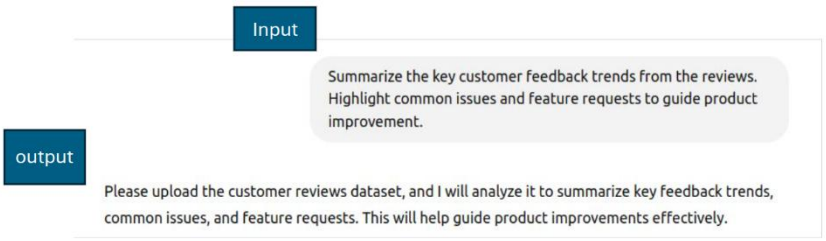
		their medical history and current health status."
Discover How GPT Transforms Education	Demonstration of GPT in personalized learning, automated grading, and knowledge dissemination in education.	**Task**: Generate a set of quiz questions on a topic for adaptive learning. **Prompt**: "Create a quiz with 5 multiple-choice questions on climate change, with increasing difficulty levels and immediate feedback for each question."
Discover How GPT Transforms Creative Industries	Explore GPT's capabilities in content creation, scriptwriting, and design ideation in the creative sectors.	**Task**: Create a script for a short advertisement promoting a new AI product. **Prompt**: "Write a 30-second script for a commercial promoting a

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		new AI-powered health app, highlighting its features and benefits.”
Discover How GPT Transforms Customer Service	Demonstration of GPT’s use in enhancing customer service through automated responses and issue resolution.	**Task**: Generate an automated response for a customer's query. **Prompt**: “Write an automated response for a customer inquiring about the return policy for a defective product, offering clear instructions and next steps.”

Discover How GPT Transforms Business

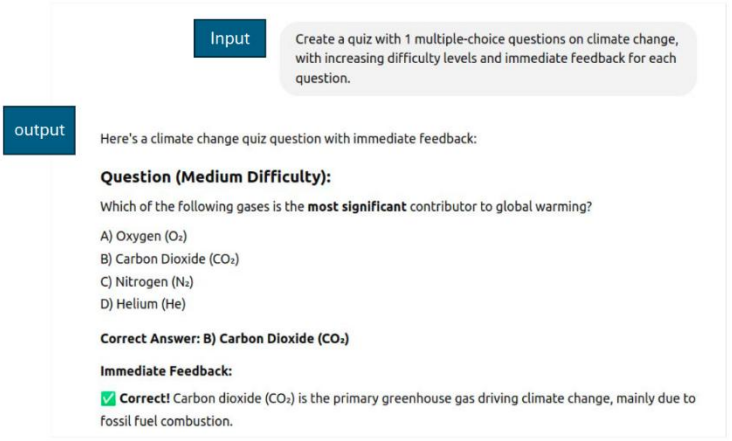
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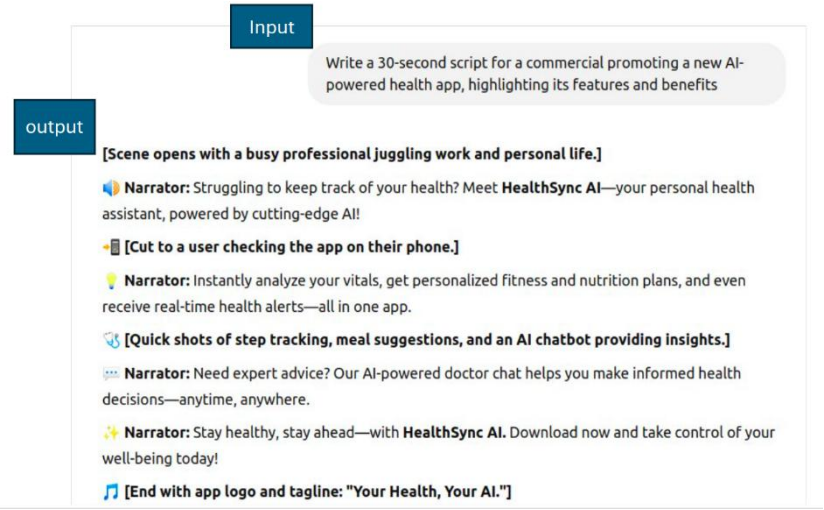
Discover How GPT Transforms Healthcare



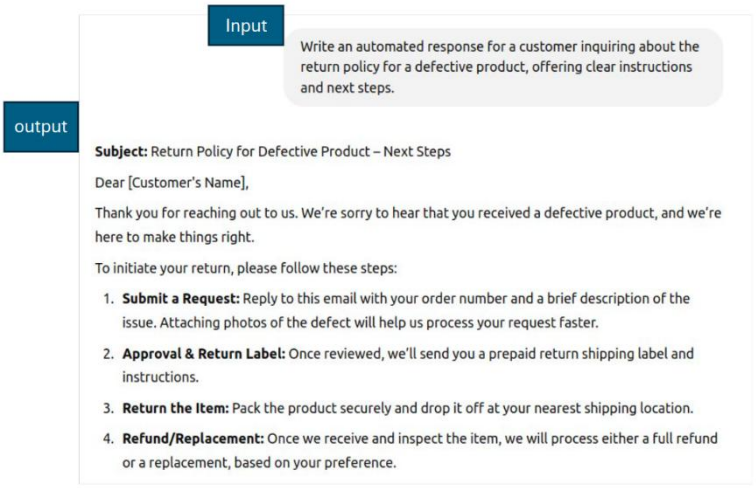
Discover How GPT Transforms Education



Discover How GPT Transforms Creative Industries



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